

Swastik Agarwala

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EDUCATION

Carnegie Mellon University

M.S. in Computer Science

Pittsburgh, PA

Expected Dec 2027

- **Current Coursework:** Computer Systems (15-513), Algorithms (15-651), Machine Learning (10-601), Distributed Systems (15-640)

Purdue University

B.S. in Computer Science, Minors in Math & Economics (GPA: 3.90/4.0, Dean's List)

West Lafayette, IN

2022 – 2026

- **Coursework:** Algorithms, Statistical Machine Learning, Deep Learning, Artificial Intelligence, Software Engineering, Operating Systems, Object-Oriented Programming, Databases, Compilers
- **Competitive Programming:** ICPC-ECNA Regional; IOI Training Camp '21; Codeforces 1862 (Expert)

TECHNICAL SKILLS

Languages: Python, Go, Java, C++, C, TypeScript, JavaScript, Scala, SQL

Backend & Systems: FastAPI, Spring Boot, Express.js, Flask, REST APIs, Microservices, Multithreading

Web & Data: React, TanStack Query, PostgreSQL, TimescaleDB, Redis, MongoDB, Docker, AWS, Grafana, Git, Linux

AI/ML: LangGraph, LangChain, LLMs, RAG, vector search, NumPy, Pandas

EXPERIENCE

Jyoti Broking Pvt. Ltd. (Proprietary Trading)

Software Developer (Full-Time)

New Delhi, India

May 2026 – Aug 2026

- Built a low-latency C++ system on a colocated exchange server that parses NSE's tick-by-tick multicast feed and reconstructs the live order book in real time (verified against exchange snapshots), with active-active redundancy and fault-tolerant gap recovery

Software Developer (Part-Time, Remote)

Jul 2024 – May 2026

- Built and operated the desk's backend: 10+ Docker services spanning real-time C++ & Go market-data ingestion (~8 MB/s across NSE, BSE, and MCX), risk/margin computation, orchestration of the full daily data pipeline, and observability (Grafana LGTM)
- Implemented a SPAN-based portfolio margin calculator; verified output matched clearing-corporation figures across 82 real client positions

AI Intern

RMgX Technologies LLP

May 2025 – Jul 2025

Remote

- Built a multi-agent travel assistant in LangGraph with coordinating sub-agents for recommendations, memory, and search, plus a multimodal RAG layer reasoning over PDFs, spreadsheets, and charts
- Cut token usage 30–40% via a dynamic context-caching memory system; built a recommendation engine using hybrid vector search (semantic + keyword) with metadata filtering

Software Engineering Intern

SML Isuzu Ltd.

May 2024 – Jul 2024

Chandigarh, India

- Shipped a React Native app for 400+ employees with QR scanning to digitize equipment-repair tracking; cut downtime 25% via a real-time notification system
- Integrated SAP with the mobile app to sync hardware specs, locations, and repair statuses in real time

Undergraduate Teaching Assistant

Purdue University

Jan 2024 – May 2026

West Lafayette, IN

- Mentored 60+ students across 3 courses in C/C++, algorithms, data structures, and systems programming

PROJECTS

Investify – Investing Learning Platform | React, Spring Boot, PostgreSQL, AWS

Aug 2024 – Dec 2024

- Built an investment-education platform with portfolio tracking, paper trading, and market analysis in a 5-person team; designed the React/TypeScript frontend with Redux Toolkit and RTK Query for state and API caching

MiniScala Compiler | Scala, C, Compilers, Systems Programming

Jan 2025 – May 2025

- Built a compiler translating MiniScala to x86 assembly (parsing, type checking, CPS translation, closure conversion, optimization passes) with a mark-and-sweep garbage collector written in C